

Data Acquisition Part 2_DataDictionary

- List variables you plan to use
- Brief description of what each variable represents and other data dictionary attributes

Target variable

loan_status: Final status of the loan (only using: Fully Paid (Non-default (0)), Charged Off (Default (1)), Default (Default (1)))

1. Fully Paid (Treated as Non-Default (0))

- The borrower has completed all scheduled payments.
- The loan principal and interest have been fully repaid.
- This represents a successful loan outcome.

2. Current

- The loan is still active and payments are being made on time.
- The final outcome (default or full repayment) is unknown.

3. Charged Off (Treated as Default (1))

- The borrower has failed to make payments for a prolonged period.
- LendingClub considers the loan unlikely to be collected and writes it off as a loss.
- Some partial recovery may occur later.

4. Late (31–120 days)

- Borrower is behind on payments by 31 to 120 days.
- High risk of eventual default.

5. In Grace Period

- Borrower has missed a payment but is still within a short grace window.
- Loan may return to current status if payment is made soon.

6. Late (16–30 days)

- Borrower is 16 to 30 days overdue on payment.
- Early stage delinquency.

7. Does not meet the credit policy. Status: Fully Paid

- Loan did not meet LendingClub's internal credit standards at origination.
- However, borrower ultimately repaid the loan successfully.

8. Does not meet the credit policy. Status: Charged Off

- Loan failed LendingClub credit policy but was issued anyway.
- Eventually defaulted.

9. Default (Treated as Default (1))

- Borrower has officially defaulted but the loan has not yet been fully charged off.
- Very rare in this dataset (only 40 records).

Borrower Demographics & Financial Capacity

annual_inc: Borrower's self-reported annual income (Numeric)

emp_length: Length of borrower's employment (Categorical)

home_ownership: Home ownership status ((Categorical (RENT, OWN, MORTGAGE)))

verification_status: Whether income was verified (Categorical (Verified / Not Verified))

Credit History & Creditworthiness

fico_range_low: Lower bound of borrower's FICO score range (Numeric)

fico_range_high: Upper bound of borrower's FICO score range (Numeric)

earliest_cr_line: Date of borrower's earliest credit line (Used to derive credit history length)

open_acc: Number of open credit lines (Numeric)

total_acc: Total number of credit lines (Numeric)

delinq_2yrs: Number of delinquencies in past 2 years (Numeric)

pub_rec: Number of derogatory public records (Numeric)

Debt Burden & Repayment Capacity

dti: Debt-to-income ratio (Key risk indicator)

revol_bal: Total revolving credit balance (Numeric)

revol_util: Revolving credit utilization rate (%) (Numeric, may need cleaning)

Loan Characteristics (Known at Origination)

loan_id(id): a unique identifier provided by LendingClub

loan_amnt: Amount of loan funded (Numeric)

term: Loan repayment term (Categorical (36 or 60 months))

int_rate: Interest rate on the loan (Numeric)
installment: Monthly payment amount (Derived from loan terms)
purpose: Purpose of the loan (Categorical (e.g., debt consolidation))
grade: LendingClub assigned loan grade (Ordinal (A–G))
sub_grade: More granular loan grade (Ordinal (e.g., A1–G5))

Temporal / Control Variables

issue_d: Month the loan was issued (Used for time-based train/test split)
addr_state: Borrower's state (Optional; may raise fairness concerns)